

AgriSentinel — PPT Speaker Script

SIH Hackathon 2026 | Problem Statement SIH260487

Project: Digital Farm Management Portal for Implementing Biosecurity Measures in Pig and Poultry Farms

Team Platform: AgriSentinel — Digital Farm Biosecurity Platform

Suggested presentation time: 8–10 minutes (+ 2 minutes Q&A)

Live Demo (use during presentation)

Resource	URL
Website	https://farm-2wx5.vercel.app/
Backend API	https://agrisentinel-api.onrender.com
API Docs	https://agrisentinel-api.onrender.com/docs
GitHub	https://github.com/ullaspn70-dotcom/FARM

Demo logins

Role	Email	Password
Farmer	farmer@bioshield.local	farmer123
Veterinarian	vet@bioshield.local	vet123
Government Officer	officer@bioshield.local	officer123

SLIDE 1 — Title

On screen: AgriSentinel | Digital Farm Biosecurity Platform | SIH260487

SPEAK (30 seconds):

Good morning / afternoon, judges. We are [Team Name], and we present **AgriSentinel** — a Digital Farm Biosecurity Platform for pig and poultry farms.

Our problem statement is **SIH260487**: building a digital farm management portal to implement biosecurity measures and protect India's livestock sector from disease outbreaks.

SLIDE 2 — The Problem

On screen: Why biosecurity matters | Outbreak risk | Manual record-keeping gaps

SPEAK (45 seconds):

Pig and poultry farms face constant biosecurity threats — sudden mortality, respiratory disease, feed contamination, and perimeter breaches.

Today, many farms still rely on **paper records**, phone calls, and delayed veterinary response. When an outbreak starts, information travels slowly. Corrective actions are not tracked. Government officers lack a real-time district

view.

The result: faster spread, economic loss, and food supply risk.

SLIDE 3 — Our Solution

On screen: AgriSentinel — one platform, three roles, closed-loop workflow

SPEAK (40 seconds):

AgriSentinel is a full-stack web platform that connects **farmers, veterinarians, and government officers** on one system.

Farmers monitor biosecurity and report incidents. Veterinarians verify reports with evidence. Officers oversee the district on a GIS map.

Every action updates a **biosecurity score** — so compliance is measurable, not guesswork.

SLIDE 4 — Problem Statement Alignment

On screen: SIH260487 requirements ↔ AgriSentinel features (table)

SPEAK (45 seconds):

We mapped every SIH requirement to a live feature:

- **Digital farm management** → multi-farm portal for poultry, pig, and mixed farms
 - **Biosecurity monitoring** → daily checklist, digital passport, risk score
 - **Incident reporting** → form with photo evidence and veterinary verification
 - **Risk assessment** → rule-based engine with score history
 - **Multi-stakeholder access** → Farmer, Vet, and Officer portals
 - **Regional oversight** → officer dashboard and GIS map
 - **Closed-loop response** → incident → verify → corrective action → score update
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SLIDE 5 — Architecture

On screen: React frontend → FastAPI backend → PostgreSQL | Vercel + Render

SPEAK (50 seconds):

Our architecture is production-grade, not a mock-up.

Frontend: React 19, TypeScript, Vite — hosted on **Vercel**

Backend: FastAPI with **40+ REST APIs** — hosted on **Render**

Database: PostgreSQL with Alembic migrations

Security: JWT authentication with role-based access

The system is **live and deployed** — you can open the demo URL right now.

SLIDE 6 — Farmer Portal

On screen: Dashboard screenshot | Biosecurity score | Checklist

SPEAK (45 seconds):

[LIVE DEMO — switch to Farmer role]

*This is the farmer dashboard. Each farm shows a **biosecurity score out of 100** and a risk level — Safe, Caution, or Critical.*

*Farmers complete a **daily biosecurity checklist** — sanitation, disinfection, visitor logs.*

*They can report health incidents, view corrective actions, and open their **Digital Biosecurity Passport** with QR code for field inspections.*

SLIDE 7 — Incident Reporting (Live Demo)

On screen: Incident form | Evidence upload

SPEAK (60 seconds):

[LIVE DEMO — Report Incident]

*When a farmer sees unusual mortality or symptoms, they open **Report Incident**.*

They select incident type, animal species, number affected, farm zone, and description.

*Critically, they can **upload photo or document evidence** — this goes directly to the veterinarian's queue.*

*The incident status is **Reported** until a vet verifies it — we never mark an outbreak as confirmed without veterinary review.*

SLIDE 8 — Veterinarian Portal

On screen: Vet verification queue | Validate / Reject / Request info

SPEAK (60 seconds):

[LIVE DEMO — switch to Veterinarian role]

*The veterinarian sees a **District Verification Queue**. Pending incidents appear at the top.*

*The vet reviews farm details, symptoms, and **uploaded evidence**.*

*They can **Validate** the incident — which auto-generates corrective actions — **Reject** a false alarm, or **Request more information** from the farmer.*

This creates accountability: every outbreak claim is professionally reviewed before district action.

SLIDE 9 — Corrective Actions & Evidence

On screen: Action list | Evidence upload | Vet action plan

SPEAK (45 seconds):

*After verification, **corrective actions** are assigned to the farmer — for example, disinfection, quarantine, or perimeter repair.*

*The farmer uploads **compliance evidence** — photos with GPS and timestamp metadata.*

Veterinarians can send a structured **Action Plan**, and our system supports **AI-assisted evidence analysis** to flag unrelated uploads.

SLIDE 10 — Government Officer & GIS

On screen: Officer dashboard | GIS map with farm nodes

SPEAK (45 seconds):

[LIVE DEMO — switch to Government Officer role]

The officer sees **district statistics** — total farms, high-risk count, open incidents, pending inspections.

The **GIS Farm Map** shows every registered farm with risk colour coding, incident markers, and spatial risk context.

This supports **data-driven field inspections** — officers prioritize high-risk farms first.

SLIDE 11 — Offline-First & PWA (Differentiator)

On screen: Offline save → auto sync | IndexedDB | Service worker

SPEAK (45 seconds):

A key differentiator: AgriSentinel works in **rural low-connectivity areas**.

Farmers can complete checklists, report incidents, and queue evidence **offline**. Data is stored safely on the device.

When internet returns, changes **sync automatically** — with duplicate protection and clear local vs server status.

This is a real offline implementation — not just an "offline label."

SLIDE 12 — Closed-Loop Workflow

On screen: Flow diagram (Registration → Score → Incident → Verify → Actions → Update)

SPEAK (40 seconds):

AgriSentinel is a **closed loop**:

1. Farm registration and biosecurity assessment
2. Continuous monitoring via checklist and risk engine
3. Incident reported with evidence
4. Veterinary verification
5. Corrective actions generated
6. Farmer submits compliance proof
7. Biosecurity score updated
8. Government oversight via GIS

No step is isolated — the full outbreak response chain is connected.

SLIDE 13 — Technology & Scalability

On screen: Tech stack table | Future roadmap

SPEAK (40 seconds):

Built today: React, FastAPI, PostgreSQL, JWT, Vercel, Render, GitHub CI/CD.

Ready to extend: Health records API, spatial risk API, AI image diagnosis, IoT sensor feeds, SMS alerts, and real state-wide farm database import.

Our modular API design lets AI and IoT teams plug in without rebuilding the platform.

SLIDE 14 — Impact & Key Differentiators

On screen: Bullet list of differentiators

SPEAK (40 seconds):

Why AgriSentinel stands out:

- Fully **cloud-deployed** live demo — not slides-only
- **Three real roles** sharing one database
- **Evidence-based** incident workflow
- **Offline-capable** for rural farmers
- **Closed-loop** biosecurity design
- Directly aligned with **SIH260487**

Impact: Faster outbreak detection, accountable veterinary response, measurable compliance, and district-level visibility for government.

SLIDE 15 — Thank You / Q&A

On screen: AgriSentinel — Protecting Farms, Securing Food Supply | Demo URL | GitHub

SPEAK (20 seconds):

Thank you, judges. AgriSentinel — **Protecting Farms, Securing Food Supply**.

Live demo: **`farm-2wx5.vercel.app`**

We welcome your questions.

QUICK 3-MINUTE VERSION (if time is limited)

Step	Time	Action
1	30s	Problem + solution (Slides 2–3)
2	45s	Farmer dashboard + report incident
3	45s	Vet verify incident
4	30s	Officer GIS map
5	30s	Architecture + thank you

JUDGE Q&A — PREPARED ANSWERS

Q: Is this only for demo or production-ready?

A: Core workflows are live on cloud with real database, authentication, file upload, and role-based APIs. AI diagnosis and IoT are on the roadmap.

Q: How is risk score calculated?

A: Rule-based engine using incidents, checklist compliance, mortality signals, and verification outcomes — recalculated when farm data changes.

Q: What if internet is poor?

A: Offline-first PWA — farmers save locally; sync runs automatically when connectivity returns.

Q: How do you prevent fake incident reports?

A: Veterinary verification is mandatory before an incident is confirmed. Evidence is reviewed. Officers see district-wide patterns on GIS.

Q: Database and security?

A: PostgreSQL on Render, JWT auth, role-based API access, HTTPS on Vercel and Render.

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